Sukh Sandhu Defense Discussion and Q&A; Guide

Detailed preparation notes for the Professional Industry Defense presentation

Core defense stance
This project is a responsible AI system design for healthcare operations triage. It supports internal routing and human review. It does not diagnose, treat, or finalize patient-facing decisions.

1. How to use this guide

Read this once before rehearsal, then use the tables as a checklist. The aim is not to memorize every line. The aim is to explain the project simply, answer questions calmly, and avoid overclaiming.

- Use the slide-by-slide section to rehearse the 15-minute flow.
- Use the Q&A bank to prepare for likely mentor questions.
- Use the evidence map to point to files when asked how the work was produced or verified.

2. Fifteen-minute timing plan

Time	Focus	What to say
0:00-1:00	Opening	State the industry, problem, and safety boundary.
1:00-3:30	Context	Explain the workflow pain in plain English.
3:30-6:00	System overview	Describe the six integrated AI layers.
6:00-8:00	Integration	Name prior projects and their contributions.
8:00-10:30	Technical decisions	Explain transparent routing and human approval.
10:30-12:30	Evaluation	Discuss four scenario outputs.
12:30-14:00	Ethics and limits	Name risks, safeguards, and prototype limits.
14:00-15:00	Close	Summarize readiness and next steps.

3. Slide-by-slide talk track

Slide	Purpose	Talk track
1	Opening	Frame the system as healthcare operations support, not diagnosis.
2	Defense requirements	Show that context, synthesis, evaluation, ethics, and reflection will all be covered.
3	Industry context	Explain healthcare operations pressure and why human oversight matters.
4	Plain-English problem	Describe before and after the system in simple terms.
5	System overview	Name the six integrated AI layers and their limited roles.
6	Case journey	Explain inputs, scoring, safety boundary, and routing output.

Slide	Purpose	Talk track
7	Prior project integration	Show how prior capstone projects combine into one workflow.
8	Technical decisions	Defend interpretability, thresholds, reproducibility, and human approval.
9	Evaluation results	Walk through the four scenario risk scores and decisions.
10	Decision rules	Make the routing logic auditable and transparent.
11	Ethics	Connect each risk to an actual safeguard.
12	Limitations	Be honest about prototype scope and validation gaps.
13	Code/run process	Explain how the code was prepared and outputs were generated.
14	Evidence package	Tell an auditor which file proves which requirement.
15	Discussion prompts	Prepare for mentor questions with a calm answer pattern.
16	Auditor checklist	Show how a layperson can inspect the package.
17	File map	Clarify the defense deck, discussion guide, auditor guide, and submission zip.
18	Close	Finish with controlled integration plus professional judgment.

4. Strong opening script

My capstone project focuses on healthcare operations, specifically the problem of routing cases when data quality, model confidence, visual evidence, and safety boundaries all matter. The system is not a diagnostic tool. It is a controlled internal triage framework that combines evidence from several AI methods and routes cases to standard queue, data quality review, human approval, or expedited review.

The important design choice is that automation is bounded. Patient-facing decisions stay with humans. The AI system helps organize evidence and make review pathways more consistent, transparent, and auditable.

5. Mentor Q&A; bank

Question	Suggested answer
Why this industry?	Healthcare operations has meaningful risk, but operations support is safer and more realistic than autonomous clinical decision-making.
What does the system do?	It combines data-quality risk, model confidence, image uncertainty, generative outlier flags, and safety boundaries to recommend a routing pathway.
Which prior projects were integrated?	Data workflow, statistical analysis, applied ML, deep learning, generative AI, and agentic workflows.
How is it more than separate projects?	The components feed one decision flow. Each signal affects the integrated routing decision.
Most important technical decision?	Interpretable routing with human approval boundaries instead of autonomous action.
What evidence shows behavior?	The scenario JSON includes four test cases with risk scores and routing decisions.
Why is SYN-003 human approval at 0.34?	Because patient-facing cases require human approval. The safety boundary overrides the numeric score.
Main limitation?	It is a prototype using illustrative thresholds and benchmark-style evidence, not clinically validated production data.
Main ethical risk?	Automation bias. The design responds with human review and visible decision traces.
Next step?	Use real domain data, calibrate thresholds, test fairness, add monitoring, and validate with experts.

6. Hard questions and calm answers

Challenge	Calm answer
This seems simple. Where is the AI?	The project demonstrates integration: statistical quality, supervised confidence, visual status, generative outlier behavior, and agentic routing.
Why not let the agent decide automatically?	The domain has safety and accountability concerns, so the system is intentionally semi-autonomous.
How do you know thresholds are correct?	The thresholds demonstrate the design pattern. Production use would require domain calibration.
Could this introduce bias?	Yes. That is why data-quality checks, human review, fairness testing, and monitoring are required.

7. Closing script

The strongest part of this project is controlled integration. I combined methods from multiple capstone projects into a single healthcare operations workflow, kept the routing logic visible, and designed safeguards around human approval and auditability. The prototype is not a production medical system, but it demonstrates the professional judgment required to build AI systems responsibly.

8. Rehearsal checklist

- Explain the project in one sentence without jargon.
- Explain why the project is not diagnostic.
- Name at least three prior projects and their contributions.
- Explain each scenario decision.
- Name one limitation and one ethical safeguard.
- Finish within 15 minutes.